

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Operations Research	Course Code:	MT 4031
	Degree Program:	BSCS	Semester:	Fall 2025
	Exam Duration:		Total Marks:	50
	Submission Date:	12-09-2025	Weight	
	Section:	All Sections	Page(s):	
	Exam Type:	Assignment-I		

Reference book: Hamdy A. Taha, Operations Research, An Introduction (10th Edition)

Instruction:

- Clearly write your name, roll no, section, Course title and assignment title on the first page.
- Write question no. as given in Assignment.
- Late submission will have no credit.
- The question that you need to be solved without GEOGEBRA, Excel and/or Python, should be solved by hand.
- Python code must be commented and executable.
- Excel Solver screenshots must include constraints, objective, and results.
- Submit your work as a single PDF file.

Questions 1:

- a. Formulate the following as a linear programming problem, find the optimal solution using **GEOGEBRA**.

A small artisan bakery makes **sourdough loaves** (x_1) and **blueberry muffins** (x_2). The bakery wants to choose daily production to maximize profit while meeting resource limits and a supplier contract. Profit per loaf is \$6 and profit per muffin is \$4. Each loaf needs 0.5 hours, each muffin 0.2 hours; at most 8 hours of oven time per day are available. Flour batch limits production so that at least 10 combined items must be produced to satisfy a minimum daily order. A fixed combinational shift requires exactly 12 hours of mixing time per day where loaf mixing takes 1 hour and muffin mixing takes 0.5 hour. The front-case can hold at most 14 items in total at any time.

Question 2:

- a. Use **Excel solver** to solve the following LP problems.

[5+5]

$$\begin{aligned}
 &\text{subject to} \\
 &\quad \text{Max } z = 6x_1 + 9x_2 + 11x_3 + 3x_4 \\
 &\quad x_1 + x_2 + x_3 + x_4 = 10,000 \\
 &\quad 70x_1 + 100x_2 + 94x_3 + 113x_4 \geq 870,000 \\
 &\quad 12x_1 + 7x_2 + 5x_3 + 18x_4 \leq 78,000 \\
 &\quad 300x_1 + 30x_2 + 5x_3 \leq 100,000 \\
 &\quad x_1 \leq 4,000 \\
 &\quad x_4 \geq 1,500 \\
 &\quad x_1, x_2, x_3, x_4 \geq 0
 \end{aligned}$$

- b. A factory produces three types of products: A, B, and C. Each product requires different amounts of two resources: labor and raw material. The goal is to determine the number of units of each product to produce that will maximize profit. The company faces following restrictions on available resources:

- Each unit of product A requires 2 hours of labor and 4 kg of raw material, and yields a profit of \$10.
- Each unit of product B requires 3 hours of labor and 2 kg of raw material, and yields a profit of \$12.
- Each unit of product C requires 1 hour of labor and 5 kg of raw material, and yields a profit of \$8.

The available resources are:

- At most 100 hours of labor per day.
- At most 120 kg of raw material per day.
- Additionally, due to market demand:
- The total number of units produced for products A and B combined must be at least 50.
- The number of units of product C must be at least 20.

Formulate the linear model. Use Excel solver to find the solution.

Question 3:

- Solve the following LLP using simplex method. Identify the optimal solution and the corresponding value of z .
- Determine which constraints are binding and which one is redundant.
- Verify your result graphically by sketching the feasible region and objective function.
- Trace the path of simplex algorithm on graphical solution space.
- Implement the simplex algorithm in **Python**.

[10]

$$\begin{aligned}
 &\text{subject to} \\
 &\quad \text{Max } z = 4x_1 + 3x_2 \\
 &\quad x_1 + x_2 \leq 8 \\
 &\quad 2x_1 + x_2 \leq 10 \\
 &\quad x_1 + 3x_2 \leq 15 \\
 &\quad 2x_1 + 2x_2 \leq 20 \\
 &\quad x_1, x_2 \geq 0.
 \end{aligned}$$

Question 4:

- a. Find the optimal solution of the following LPP using the big M method and two phase technique. (manual solution)
- b. Implement in **Python** using the **Big M algorithm** and the **Two phase method**. [10+5]

$$\text{Max } z = 5x_1 + 4x_2$$

subject to

$$x_1 + x_2 = 6$$

$$2x_1 + x_2 \geq 8$$

$$x_1 + 3x_2 \leq 15$$

$$x_1, x_2 \geq 0.$$

Question 5:

[10]

Consider the following LP model,

$$\text{Max } Z = 5x_1 + 7x_2$$

subject to

$$x_1 \leq 6$$

$$2x_1 + 3x_2 \leq 19$$

$$x_1 + x_2 \leq 8$$

$$x_1, x_2 \geq 0.$$

Solve it graphically and

- a. Find the range of optimality.
- b. Suppose that the unit revenue of product 1 is fixed at its current value $c_1 = 5$, what is the associated optimality range for the unit revenue for product 2, that will keep the optimum unchanged?
- c. What is the dual price of constraints 2 and 3?